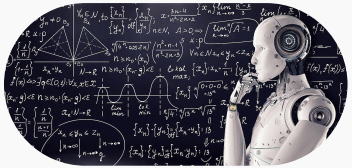
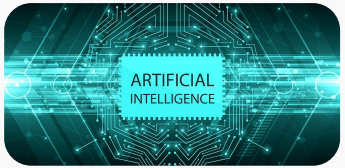
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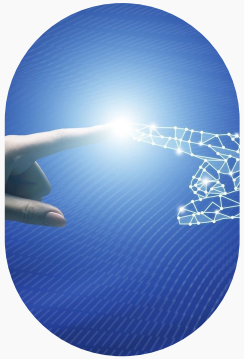
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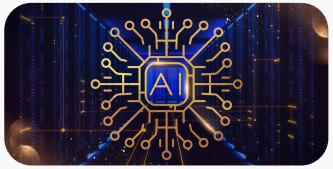
Problem Spaces, and Search

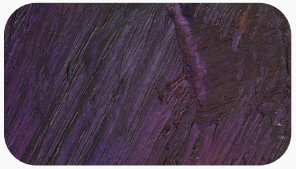
Heuristic Search Techniques

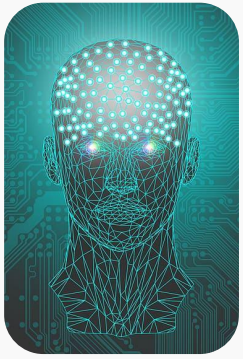












AGENDA

Introduction: The AI Problems, Underlying Assumption, What is Artificial Intelligence? AI Technique, Level of the Model.

Problem Spaces and Search: Defining the Problem as a State Space Search, Production Systems, Problem Characteristics, Production System Characteristics, and Issues in the Design of Search Programs.

Heuristic Search Techniques: Generate-and-Test, Hill Climbing, Best-first Search, Problem Reduction, Constraint Satisfaction, Means-ends

DEFINITION

Artificial means **made by humans** instead of occurring naturally.

Intelligence is the **ability to learn, think, understand, solve problems, and make decisions**.

Artificial Intelligence (AI) is the ability of a machine to think and learn like a human.

AI is the intelligence shown by machines.

What is AI? - Science of making machines that think and act intelligently

Who was John McCarthy?

- Known as the **Father of Artificial Intelligence**.
- Coined the term **"Artificial Intelligence"** in **1956**.
- Organized the **Dartmouth Conference(held in the United States during the summer of 1956.)**, where AI became an official field of study.

"Artificial intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs."
— John McCarthy (1956)

APPLICATIONS OF AI

Artificial Intelligence (AI) is used in many areas to make work faster, easier, and more accurate.

In healthcare, AI helps doctors diagnose diseases and recommend treatments. In education, it provides personalized learning and smart tutoring systems. Banks use AI to detect fraud and improve customer service. In transportation, AI powers self-driving cars and traffic management systems. Industries use AI for automation and quality control, while retailers use it for product recommendations and inventory management. AI also supports agriculture through crop monitoring and smart irrigation, improves cybersecurity by detecting threats, enhances entertainment with content recommendations, enables smart homes, assists businesses in marketing, and helps protect the environment through climate monitoring and resource management. Overall, AI is transforming industries by increasing efficiency, reducing human effort, and improving decision-making.

APPLICATIONS OF AI

1. HEALTHCARE



AI helps in disease detection, diagnosis, personalized treatment and drug discovery.

2. FINANCE



AI is used for fraud detection, algorithmic trading, risk assessment and customer support.

3. EDUCATION



AI enables personalized learning, smart tutoring systems and automated grading.

4. TRANSPORTATION



AI powers self-driving vehicles, route optimization and traffic management.

5. MANUFACTURING



AI is used for predictive maintenance, quality control and process automation.

6. RETAIL



AI enhances customer experience, demand forecasting, inventory management and recommendations.

7. AGRICULTURE



AI helps in crop health monitoring, yield prediction, pest control and smart irrigation.

8. ENTERTAINMENT



AI powers content recommendations, video editing, game AI and virtual characters.

9. SMART HOMES



AI enables smart assistants, energy management, security and home automation.

10. CYBERSECURITY



AI detects threats, prevents attacks, analyzes patterns and strengthens system security.

11. MARKETING



AI is used for customer segmentation, targeted ads, sentiment analysis and campaign optimization.

12. ENVIRONMENT



AI helps in climate modeling, pollution monitoring, wildlife protection and resource management.



AI is transforming every industry by improving efficiency, accuracy and innovation, making our lives easier and better.











Real-Life AI Examples

- ChatGPT – Conversational AI
- Google Assistant – Voice assistant
- Amazon Alexa – Smart home assistant
- Netflix – Recommendation system
- Tesla – Autonomous driving features

EXAMPLE OF AI



Identify 10 AI applications that you use in your daily life.
Briefly explain how AI helps in each application.



Paper Title: *Computing Machinery and Intelligence* (1950)

Author: Alan Turing

Importance: Introduced the question **"Can Machines Think?"** and the **Turing Test**, laying the foundation of artificial intelligence.

A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

📖 Recommended Research Paper

History of Artificial Intelligence (AI)

1943 – First AI Idea

- Scientists **Warren McCulloch** and **Walter Pitts** proposed the first mathematical model of an artificial neuron.
- This became the foundation of neural networks.

1950 – Can Machines Think?

- Alan Turing published the paper **"Computing Machinery and Intelligence."**
- He introduced the famous **Turing Test** to check whether a machine can think like a human.

1956 – Birth of AI

- John McCarthy coined the term **"Artificial Intelligence"** at the Dartmouth Summer Research Project on Artificial Intelligence.
- This event is considered the **birth of AI** as a field.

1980s – Expert Systems

- AI was used to create **Expert Systems (Bank Loan Approval System - Decides whether a loan should be approved, Crop Advisory System - Helps farmers with crop and fertilizer recommendations**.)
- These systems helped **doctors, engineers, and businesses** solve problems and make decisions.

1997 – AI Wins in Chess

- **Deep Blue**, an AI computer, defeated world chess champion **Garry Kasparov**.
- It showed that AI could solve complex problems.

2010s – Deep Learning

- AI became **smarter and more accurate**.
- It learned from **large amounts of data** and improved its performance.

2022 – Generative AI

- **ChatGPT** made **Generative AI** popular.
- It can **write text, answer questions, generate code, and create content** from user prompts.

FUTURE
& AI

What is Generative AI?

Definition:
Generative AI creates new content such as text, images, music, code, or videos based on the user's input (prompt).

How it works:

- You give a prompt.
- AI generates a response.
- It only provides information or suggestions.
- The final decision and action are taken by the user.

Example:

Prompt: "Plan a 7-day Europe trip."

Generative AI Response:

- Day 1: Visit Paris
- Day 2: Explore the Eiffel Tower
- Day 3: Travel to Switzerland
- Day 4: Visit the Alps
- Day 5: Explore Italy
- Day 6: Shopping
- Day 7: Return home

The AI **only suggests the plan**. It **does not book flights or hotels**.

FUTURE & AI

What is Agentic AI?

Definition:
Agentic AI is an intelligent AI system that can **plan, make decisions, use different tools, and complete tasks automatically** with very little human help.

How it works:

- Understands your goal.
- Plans the steps.
- Uses different applications or websites.
- Performs tasks automatically.
- Updates the plan if something changes.

Example:

You say,

"Plan my 7-day Europe trip with a budget of ₹2 lakh."

Agentic AI will:

- Search for flights
- Compare hotel prices
- Book hotels
- Reserve attraction tickets
- Suggest restaurants
- Monitor weather
- Change your schedule if a flight is delayed

It acts like your **personal travel assistant**.

Q: Who asked the question **"Can machines think?"**
A: Alan Turing

Q: Who coined the term **"Artificial Intelligence"?**
A: John McCarthy

Q: Who is known as the **Father of Artificial Intelligence**?
A: John McCarthy

Q: In which year was the term **Artificial Intelligence** introduced?
A: 1956

QUIZ

QUIZ

Q1. Why is Alan Turing important in the history of AI?

Answer:

- Alan Turing asked the question **"Can machines think?"** in 1950.
- He introduced the **Turing Test** to check whether a machine can behave like a human.
- He laid the **foundation of Artificial Intelligence**.

Q2. Why is John McCarthy called the Father of AI?

Answer:

- John McCarthy coined the term **"Artificial Intelligence"** in 1956.
- He organized the **Dartmouth Conference**, where AI became an official field of study.
- He defined AI as the **science and engineering of making intelligent machines**.
- Therefore, he is known as the **Father of Artificial Intelligence**.

The Turing Test is a test used to determine whether a machine can behave like a human during a conversation.

Human + AI → Conversation → Judge cannot identify the AI → AI passes the Turing Test.

How Does It Work?

1. A **person** chats with **two participants**.
2. One participant is a **human** and the other is an **AI machine**.
3. The person **does not know** who is the human and who is the AI.
4. If the person **cannot correctly identify the AI**, then the AI is said to **pass the Turing Test**.

TURING TEST

What is the Turing Test?

Definition:
The **Turing Test** is a test proposed by **Alan Turing** in **1950** to determine whether a machine can **think or behave like a human** during a conversation.

Simple Explanation

In the Turing Test:

- A **human judge** chats with **both a human and a machine**.
- The judge does **not know** which one is the machine.
- If the judge **cannot reliably tell the difference**, the machine is said to have **passed the Turing Test**.

Example

Imagine you are chatting online with two people:

- **Person A** – A human
- **Person B** – An AI chatbot

If you cannot identify which one is the AI because its answers are very human-like, then the AI has **passed the Turing Test**.

"Which is harder for a computer — playing chess, or recognizing your mother's voice on a phone call?"

Actually, recognizing a voice is much harder for a computer! Today we'll understand why."

Task	Difficulty for AI	Why?
Playing Chess/Tic-Tac-Toe	Medium	Rules are fixed, moves are limited, and the environment is predictable.
Recognizing Voice over a Phone Call	Very Hard	People have different accents, speak at different speeds, there may be background noise, poor network quality, and multiple languages.

"AI problems = Tasks that are EASY for humans but HARD for computers"

Mundane Tasks

- Perception (Vision, Speech)
- Natural Language (Understanding, Generation, Translation)
- Commonsense reasoning
- Robot control

Formal Tasks

- Games (Chess, Backgammon, Checkers, Go)
- Mathematics (Geometry, Logic, Integral calculus, Proving properties of programs)

Expert Tasks

- Engineering (Design, Fault finding, Manufacturing planning)
- Scientific analysis
- Medical diagnosis
- Financial analysis

Some of the Task Domains of Artificial Intelligence

Type	Meaning	Examples	Why Hard for Computers?
Mundane Tasks	Everyday things we do without thinking	Vision, Speech, Natural Language, Robot Control	Noisy signals, needs world knowledge, too many things happening at once
Formal Tasks	Things that LOOK smart/hard	Chess, Maths, Logic, Theorem Proving	Combinatorial explosion — too many possibilities to check
Expert Tasks	Things only trained professionals can do	Medical Diagnosis, Engineering Design, Financial Analysis	Needs deep, specialized human knowledge

Table: Some of the Task Domains of Artificial Intelligence / Types of AI Problems

QUESTION: Why is Voice Recognition Harder?

- Different accents
- Different languages
- Fast or slow speech
- Background noise
- Poor phone signal
- Similar-sounding words
- Understanding the meaning of the sentence (context)

The AI Problems

An **AI problem** is a problem that requires **thinking, reasoning, learning, or decision-making** to find the best solution.

Some problems cannot be solved by following fixed steps. AI helps by **thinking, analyzing different options, and choosing the best solution**.

Examples

1. Google Maps

Problem: Finding the fastest route from your home to college.

- AI analyzes roads, traffic, and distance before suggesting the best route.

2. Face Unlock

Problem: Identifying the correct person.

- AI compares your face with the stored image and unlocks the phone.

3. Google Translate

Problem: Translating one language into another.

- AI understands the meaning of a sentence before translating it.

4. Chess Game

Problem: Finding the best move.

- AI analyzes many possible moves before choosing the best one.

5. Medical Diagnosis

Problem: Detecting diseases.

- AI studies medical images and helps doctors identify diseases.



Identify the AI Problem

Classify each scenario into the appropriate AI problem.

Scenario	AI Problem
Finding the shortest path to college	Search Problem
Solving Sudoku	Constraint Satisfaction
Detecting spam emails	Machine Learning
Face Unlock on a phone	Computer Vision
Google Translate	Natural Language Processing
Alexa understanding voice commands	Speech Recognition
Amazon recommending products	Recommendation System
Robot delivering food in a hospital	Planning and Robotics

Underlying Assumption

In **1976**, AI researchers **Allen Newell** and **Herbert Simon** proposed an idea called the **Physical Symbol System Hypothesis**.

They believed:

> **"If a computer can store and process symbols like the human brain, it can perform intelligent tasks."**

This idea became the **foundation (underlying assumption)** of Artificial Intelligence.

EXAMPLE

What is **4 + 6**?

- Your **brain** processes the symbols **4**, **+**, and **6** and answers **10**.
- A **computer** also processes the same symbols and answers **10**.

Since **both the human brain and computers process symbols**, Newell and Simon believed that **computers could also perform intelligent tasks**.

A computer that can process symbols like the human brain can perform intelligent tasks. It is a **hypothesis (idea, assumption, or educated guess that has not yet been proven)**, not a proven fact.

Hypothesis = An idea that may be true, but needs to be tested.

What is a Symbol?

A **symbol** is anything that represents something.

Example

🍎 = Apple

🚗 = Car

The symbol is **not the real object**, it only represents it.

What is a Physical Symbol System?

A **Physical Symbol System** is anything that can:

- Store symbols
- Understand symbols
- Use symbols to solve problems

Underlying assumptions are the **basic ideas or beliefs** on which an AI system works.

Underlying assumptions are the **basic assumptions made while designing an AI system**. These assumptions define **how the AI represents knowledge and solves problems**.

Before solving any problem, AI **assumes certain information is true**. These assumptions help AI make decisions.

AI assumes that knowledge can be stored and used to make intelligent decisions.

Underlying Assumption

Underlying Assumptions of AI

- Intelligence can be simulated by computers.
- Knowledge can be represented inside a computer.
- Computers can use knowledge to reason and solve problems.
- Intelligent behavior can be achieved through search and problem-solving techniques.

AI assumes that a computer can **store knowledge**, **use that knowledge to think**, and **find solutions** to problems in an intelligent way.

1. Intelligence Can Be Simulated by Computers

AI assumes that a **computer can perform some intelligent tasks like a human**, such as learning, reasoning, and making decisions.

Example

- **ChatGPT** answers questions like a human.
- **Alexa** understands voice commands and responds.

2. Knowledge Can Be Represented Inside a Computer

AI assumes that **information can be stored inside a computer** in the form of facts, rules, or data.

Example

- **Google Maps** stores information about roads, traffic, and locations.
- It uses this information to guide users.

3. Computers Can Use Knowledge to Reason and Solve Problems

AI assumes that a computer can **use stored knowledge to think logically and make decisions**.

Example

- **Chess AI** uses the rules of chess and the current board position to choose the best move.
- **Medical AI** uses symptoms and medical data to help identify diseases.

4. Intelligent Behavior Can Be Achieved Through Search and Problem-Solving Techniques

AI assumes that **a computer can search through different possible solutions and select the best one**.

Example

- **Google Maps** checks many possible routes and selects the fastest one.
- A **robot** searches for the best path to reach its destination while avoiding obstacles.

AI Technique

An AI Technique is a smart way of solving problems using Search, Knowledge, and Abstraction instead of only following fixed rules.

Example

Question: How does Google Maps find the best route?

- It doesn't guess randomly.
- It **searches** different routes.
- It uses **knowledge** about roads and traffic.
- It gives the **best path**.

This is an **AI Technique**.

What is AI Technique ?

Core Idea: "Intelligence requires Knowledge"
Knowledge must be:
• Generalized • Easy to modify
• Understandable
• Usable if incomplete (AI should still work even when some information is missing.)
• Narrows options (AI should use knowledge to eliminate unlikely choices and focus on the best possible options.)

AI Technique	Simple Meaning	Everyday Example
Search	Finding the best solution by checking different options. / No direct solution exists	You lose your mobile phone and search in your room, bag, and sofa until you find it.
Use of Knowledge	Using what you already know to make the correct decision./ Understand structure	You see dark clouds and carry an umbrella because you know it may rain.
Abstraction	Focusing only on important information and ignoring unnecessary details.	While buying a laptop, you check the processor, RAM, and battery instead of its wallpaper or stickers.

1. Search

Search means looking at different options to find the best answer.

Everyday Example

Imagine you lose your **mobile phone** at home.

What do you do?

- Check the bedroom.
- Check the sofa.
- Check the kitchen.
- Check your bag.

You keep searching until you find it.

👉 **AI also searches through different possible solutions before choosing the best one.**

Examples:

- Google Maps finds the shortest route.
- Chess AI searches for the best move.

2. Use of Knowledge

Knowledge means using what you already know to make a good decision.

Everyday Example

Suppose it is cloudy, there is thunder, and strong wind.

What do you think?

👉 **It may rain.**

You did not guess randomly—you used your **knowledge** from past experience.

Similarly, AI uses stored knowledge to solve problems.

Examples:

- A doctor uses symptoms to identify a disease.
- AI recommends a movie based on your previous choices.

3. Abstraction

Meaning

Abstraction means focusing only on important information and ignoring unnecessary details.

Everyday Example

Suppose you want to buy a **laptop**.

Do you care about:

- The wallpaper? ❌
- The sticker on the laptop? ❌
- The processor, RAM, and battery? ✅

You ignore unimportant details and focus on what matters.

This is **abstraction**.

Another example:
When using **Google Maps**, you only see roads and directions—not every tree, house, or streetlight.

AI Technique	Examples
_____ ?	1. Finding the shortest route using Google Maps. 2. Searching for a file on your laptop. 3. Finding an empty parking space.
????	1. A doctor diagnosing a disease based on symptoms. 2. Google Assistant answering "What is the capital of India?" 3. A chatbot answering customer questions.
??????	1. A map shows roads, not every house or tree. 2. A student ID card contains only important details (name, photo, ID number). 3. An ATM receipt shows only transaction details, not the machine's internal process.

Identify AI Technique for each abovementioned example

"If a teacher asks two students to calculate 25 × 18, will both use exactly the same method?"

Student A:

- Multiplies on paper.

Student B:

- Uses mental math.

Student C:

- Uses a calculator.

"Did everyone get the correct answer?"

Yes.

"Did everyone think in the same way?"

No.

This is exactly what the "Level of the Model" is about.

The Level of the Model is about deciding whether an AI system should think like a human or simply solve the problem in the best possible way.

Before building any AI system, researchers must answer one question:

"Should this AI behave like a human, or should it simply perform the task as efficiently as possible?"

This decision affects **how the AI is designed**.

Level of the Model.

When we build AI, we have to decide:

Option 1

Should AI **think exactly like humans?**

OR

Option 2

Should AI **simply solve the problem correctly, even if its method is completely different?**

That is the **Level of the Model**.

There are **two goals** while building AI.

Goal	Meaning
Copy Human Thinking	AI behaves exactly like humans.
Solve the Problem	AI can use any method as long as it gives the correct answer.

Example: Navigation

You want to go from College to Home.

Human

"I know this road has traffic.
I'll take another route."

Google Maps

Checks millions of routes using algorithms.

Both reach home.

But...

Did Google Maps think like you?

No.

It used a different method.

Still...

It solved the problem.

As we move from Level 1 to Level 4, the AI model becomes more intelligent and closer to human thinking.

Level of AI Model	Definition	Example
1. Knowledge Level	Focuses on **what the AI knows** (facts and information).	A chatbot knows that **Ahmedabad is in Gujarat**.
2. Logical Level	Uses **rules and logic** to reason and make decisions.	**If it is raining, carry an umbrella.**
3. Implementation Level	Describes **how the AI is programmed** using algorithms and data structures.	A face recognition app uses a machine learning algorithm.
4. Hardware Level	The **physical devices** that run the AI system.	Computer, GPU, mobile phone, robot.

Question – "What is 5 + 3?"

Level	Example
Knowledge Level	AI knows the numbers **5** and **3**.
Logical Level	AI understands that it needs to **add** the numbers.
Implementation Level	AI performs the **addition** and gets **8**.
Hardware Level	The calculation runs on the **computer or mobile phone**.

The **Levels of AI Models** show how an AI system works step by step. They explain **what AI knows, how it thinks, how it performs a task, and where it runs**.

Levels of AI Models

Example 1: **"What is 10 × 5?"**

Level	Explanation
Knowledge Level	AI knows the numbers **10** and **5**.
Logical Level	AI understands that it should **multiply** the numbers.
Implementation Level	AI calculates the answer (**50**) using a program.
Hardware Level	The calculation runs on a computer or mobile phone.

Example 2: **"Who is the Prime Minister of India?"**

Level	Explanation
Knowledge Level	AI knows information about the Prime Minister.
Logical Level	AI understands that the user is asking about a person.
Implementation Level	AI searches its stored knowledge and gives the answer.
Hardware Level	AI runs on a mobile phone or computer.

Example 3: **"Set an alarm for 7:00 AM."**

Level	Explanation
Knowledge Level	AI knows what an alarm is and understands time.
Logical Level	AI understands the request to set an alarm for 7:00 AM.
Implementation Level	AI saves the alarm using its program.
Hardware Level	The alarm runs on your smartphone.

Example 4: **"Play my favorite song."**

Level	Explanation
Knowledge Level	AI knows your music library or playlist.
Logical Level	AI understands that you want to play a song.
Implementation Level	AI finds and starts the song.
Hardware Level	The song plays through your phone or smart speaker.

TASK

Classify the following 10 tasks into Mundane, Formal, or Expert Tasks, and justify your answer in one line each:

1. Translating a Hindi sentence to English
2. Solving a calculus integration problem
3. Diagnosing a heart condition from an ECG report
4. Recognizing your friend's voice on a call
5. Playing a game of Go
6. Designing a bridge structure
7. A robot picking up a cup without dropping it
8. Predicting stock market trends
9. Proving a geometry theorem

TASK

For each scenario below, identify whether it primarily uses Search, Use of Knowledge, or Abstraction and explain why:

1. A GPS app finds 5 alternate routes when there's traffic
2. A doctor rules out flu because the patient has no fever
3. An online map only shows highways, not every lane in a colony
4. A chess engine explores possible move sequences 10 steps ahead
5. A weather app predicts rain by analyzing cloud patterns

Pick any one daily task (e.g., finding a lost item, choosing what to wear, picking a movie to watch). Describe HOW you do it step-by-step. Then describe how a computer might solve the SAME task using a completely different method. Do both reach the same result?

X | | O

| X |

O | |

It is **X's turn**.

(Program 1)

Computer A already knows the answer because it has memorized all Tic-Tac-Toe boards. It simply copies the stored move.

Did Computer A THINK before making the move? Justify your answer.

☐ Yes

☐ No

(Program 2)

Computer B follows only one rule:

"If the opponent can win, block them. Otherwise, choose a corner."

Did Computer B use a RULE to make the move?

☐ Yes

☐ No

Which computer is smarter?
C , It thinks before moving.
It plans ahead.
It does not simply remember answers.
It uses reasoning.

Step 3 (Program 3)

Computer C looks at the whole board, thinks about future moves, and then chooses the best move.

Which computer is the smartest?

☐ Computer A

☐ Computer B

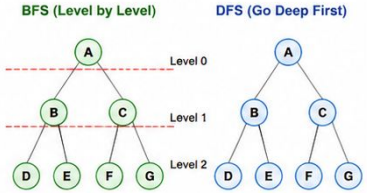
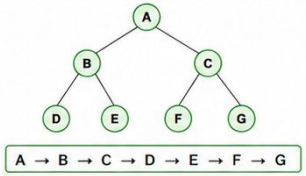
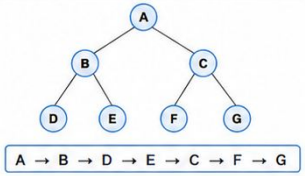
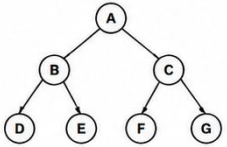
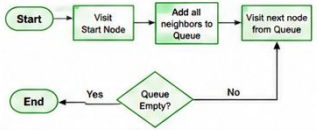
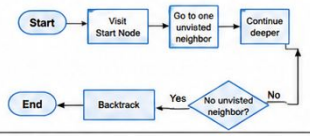
☐ Computer C

Which method is more intelligent?

☐ Memorizing all answers

☐ Using knowledge and reasoning to choose the best move

AI is not about memorizing answers. It is about using knowledge, reasoning, and planning to solve problems intelligently.

Basis	BFS (Breadth-First Search)	DFS (Depth-First Search)	Explanation / Diagram
Meaning	Explores all nodes level by level before moving to the next level.	Explores one branch as deep as possible before backtracking.	
Search Pattern	Wide (Level-wise)	Deep (Branch-wise)	
Data Structure Used	Queue (FIFO)	Stack (LIFO) or Recursion	
Working	Visits all neighboring nodes first.	Visits one path completely, then returns to explore other paths.	
Shortest Path	Yes (in unweighted graphs) ✅	Not guaranteed ❌	
Memory Usage	More (stores all nodes of a level)	Less (stores one path at a time)	
Real-Life Example	Searching for a classroom by checking every room on the current floor first, then moving to the next floor.	Searching for a classroom by entering one corridor and checking every room until the end, then returning to the next corridor.	
Traversal Example (For Tree)			
Flow Diagram			
Advantages	• Finds shortest path in unweighted graphs • Complete (will find the goal if it exists) • Good for finding nearest solution	• Uses less memory • Simple to implement (with recursion) • Good for exploring all paths	
Disadvantages	• Uses more memory • Slower when the solution is deep	• Does not guarantee shortest path • May get stuck in infinite path (in some cases) • Not complete in infinite-depth graphs	
Easy Memory Trick	**BFS = Breadth = "Check all nearby nodes first."** **Think: **WIDE SEARCH****	**DFS = Depth = "Go deep into one path first."** **Think: **DEEP SEARCH****	

Breadth- First Search (BFS) Algorithm

Algorithm

1. Start from the root (start) node.
2. Insert the start node into the **Queue**.
3. Remove the first node from the queue and visit it.
4. Add all its unvisited neighboring nodes to the queue.
5. Repeat Steps 3 and 4 until the queue becomes empty or the goal node is found.
6. Stop.

Data Structure Used

- **Queue (FIFO – First In, First Out)**

Depth-First Search (DFS) Algorithm

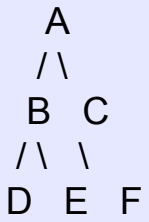
Algorithm

1. Start from the root (start) node.
2. Push the start node onto the **Stack**.
3. Pop the top node from the stack and visit it.
4. Push all its unvisited neighboring nodes onto the stack.
5. Repeat Steps 3 and 4 until the stack becomes empty or the goal node is found.
6. Stop.

Data Structure Used

- **Stack (LIFO – Last In, First Out)**

Step	Operation	Queue (Front → Rear)	Visited
1	Insert A	A	-
2	Remove A, visit A, insert B, C	B → C	A
3	Remove B, visit B, insert D, E	C → D → E	A → B
4	Remove C, visit C, insert F	D → E → F	A → B → C
5	Remove D, visit D	E → F	A → B → C → D
6	Remove E, visit E	F	A → B → C → D → E
7	Remove F, visit F	Empty	A → B → C → D → E → F

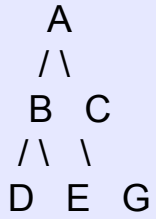


Start Node = A

BFS Solution

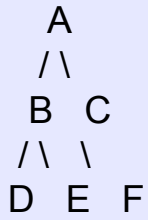
Apply BFS starting from node A to find the goal node G.

Note: Stop when the goal is reached and show the path.

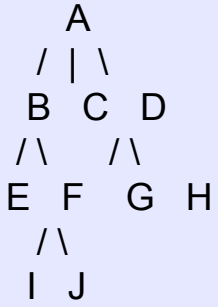




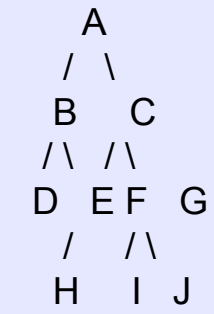
Apply BFS on the following graph and write the traversal order.
Note: Visit all nodes and write the traversal order.

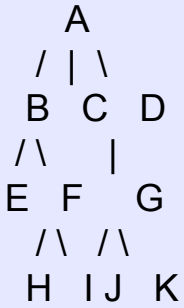


Apply Breadth-First Search (BFS) starting from node A. Write the queue contents at each step and the final traversal order.

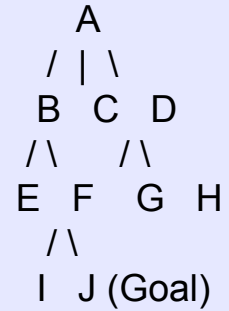


Apply the Depth-First Search (DFS) algorithm to the following tree starting from node A. Show the contents of the stack at each step and write the final DFS traversal order.



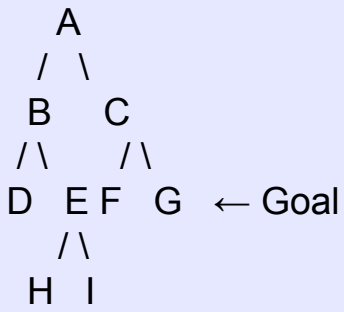


Apply the Depth-First Search (DFS) algorithm to find the goal node J starting from node A. Show the stack after each iteration and write the path followed until the goal node is reached.





Consider the following tree. Perform **Depth-First Search (DFS)** starting from node **A**. Assume the search always explores the **left child before the right child**. Find the path until the **goal node G** is reached.



Perform DFS starting from **A** (left child first). Stop when the **goal node E** is found.

